CPSC 368 – Phase 2: Formal Research Proposal

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1. Research Questions

Main Question:

How consistent are movie ratings between IMDb and MovieLens for the same movies released between 2020 and 2025?

Alignment = IMDb_avg − MovieLens_avg (per movie)

Disagreement = |Alignment| (magnitude of difference)

Sub-Questions

1. Does the average disagreement (mean |Alignment|) differ across selected genres (2020–2025)?
2. Does the average disagreement vary across release years (2020–2025)?
3. Is there a measurable correlation between user engagement (MovieLens tag counts and IMDb vote counts) and disagreement magnitude?

2. Justification of Interest

IMDb and MovieLens represent different user populations. IMDb reflects a broad public audience, while MovieLens is widely used in research contexts. Understanding cross-platform rating consistency helps determine whether ratings are interchangeable, whether disagreement is systematic, and whether engagement is associated with rating divergence.

3. Hypotheses

- H1: Most movies will exhibit small disagreement (|Alignment| close to 0).
- H2: Certain genres will exhibit higher mean disagreement than others.
- H3: Higher engagement will be positively correlated with disagreement magnitude.

4. Data Sources

Dataset 1: MovieLens (GroupLens)
URL: https://grouplens.org/datasets/movielens/

Variables used: movieId, imdbId, rating, genres, tag count.

Relevant Version (for this project):
We plan to use the **MovieLens Latest Full dataset** (`ml-latest.zip`).
This version includes:

- Approximately **33,000,000 ratings** of movies.
- Around **86,000 movies** rated.
- Around **330,975 users** who contributed ratings.
- About **2,000,000 tag applications** from users.

Scope:

Because our research focuses on movies released from **2020 to 2025**, we will filter the MovieLens data accordingly after joining with external metadata sources (IMDb and TMDB). The number of movies in this filtered subset is expected to be **significantly smaller than the full dataset**, ensuring a manageable size for analysis (target 100-2000 movies).

Dataset 2: IMDb Official Dataset
URL: https://datasets.imdbws.com/

Variables used:

`tconst`, `startYear`, `genres` (from `title.basics.tsv`)

`averageRating`, `numVotes` (from `title.ratings.tsv`)

Scope:

The IMDb official dataset contains structured metadata for millions of titles across different media types, including films, TV series, and short productions. For this project, we will use two specific files:

- `title.basics.tsv` (approximately 11+ million titles)

- `title.ratings.tsv` (approximately 1+ million rated titles)

We will restrict our analysis to titles where:

- `titleType = 'movie'`
- `startYear` is between **2020 and 2025**
- A corresponding rating exists in `title.ratings.tsv`

After filtering by title type and release year, we expect the working subset to consist of several thousand contemporary films.

This filtered subset will then be joined with MovieLens data using the common IMDb identifier (`imdbId` converted to `tconst`). The resulting merged dataset will allow us to compute:

- Cross-platform rating alignment
- Disagreement magnitude (absolute difference)
- Engagement metrics using `numVotes`

All joins and aggregations will be performed using SQL in the relational database environment.

Dataset 3: TMDB Movie Metadata (Kaggle)

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URL:
https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset-2023-930k-movies

Variables used:
`id` (TMDB movie ID),
`title`,
`release_date`,
`genres`,
`vote_average`,
`vote_count`

Scope:

The TMDB Full Movies Dataset 2024 contains metadata for approximately **900,000–1,000,000 movies**, including both historical and contemporary releases. The dataset includes structured attributes such as release dates, genre classifications, and aggregated user voting statistics.

For this project, we will:

- Extract the release year from the `release_date` field
- Filter movies released between **2020 and 2025**
- Use genre metadata to cross-validate or supplement genre classifications from IMDb
- Optionally use `vote_average` and `vote_count` as secondary engagement measures

After filtering by release year and matching by movie title and year, only a subset of contemporary films will be retained for integration with MovieLens and IMDb.

This dataset serves as an additional metadata source to improve matching accuracy and ensure consistent genre classification across platforms.

All joins and filtering will be implemented using SQL before downstream analysis.

5. Methodology

1. Aggregate MovieLens ratings per movie (AVG, COUNT).
2. Join with IMDb ratings using IMDb ID.
3. Compute Alignment and Disagreement per movie.
4. Compute mean disagreement by genre and by year.
5. Compute the correlation between disagreement and engagement measures.

All aggregation will be performed using database queries; Python will be used for visualization.

Before final filtering, we will perform exploratory data analysis, including counts of movies per genre and per release year, to ensure sufficient representation and manageable dataset size.

Disagreement values close to 0 indicate strong cross-platform agreement, while larger values indicate stronger divergence.

6. Data Combination Plan

Primary join key: IMDb ID. MovieLens links.csv maps movieId to imdbId, which is converted to IMDb tconst format. TMDB entries will be joined using IMDb ID when available; otherwise, title and release year will be used. Ambiguous matches will be excluded and documented.

Standardization steps include normalizing ID formatting, date formats (YYYY), genre labels, and handling missing values before joining.

7. Work Distribution Plan

Jiahao Li

- Design relational database schema (Oracle).
- Implement SQL queries for rating aggregation and alignment calculations.
- Perform year-based and genre-based grouping analysis.

Tim Zeng

- Data cleaning and preprocessing (ID normalization, format standardization)
- Implement MongoDB data model and NoSQL queries (Phase 4).
- Assist with dataset integration across platforms.

Loay Al Abri

- Perform exploratory data analysis.
- Compute correlation between engagement metrics and disagreement.
- Generate visualizations using Python.

All Members

- Research question refinement.
- Interpretation of results.
- Writing and editing of the final research paper.

- Presentation preparation.

8. Declaration of AI Tool Use

We used ChatGPT (OpenAI GPT-5) to review clarity of writing, verify dataset structure, and refine the wording of the research proposal.

All research design decisions, dataset selection, methodological planning, and research questions were developed independently by the group. No AI tool was used to generate database queries, perform data analysis, or complete technical implementation tasks.